

Math 548

Portfolio Assignment 9

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Problem 1 (F25, P3). Let R be an integral domain. Suppose that a and b are non-associate irreducible elements in R , and the ideal (a, b) generated by a and b is a proper ideal. Show that R is not a principal ideal domain.

Proof. Suppose there exists an element $c \in R$ such that $(a, b) = (c)$. Then $a = cr$ and $b = cs$ for some $s, r \in \mathbf{Z}$. Since a and b are irreducible, either c is a unit or s and r are units. If c is a unit, then $(a, b) = R$, contradicting the fact that (a, b) is a proper ideal. Hence it must be the case that r and s are units. This implies that $ar^{-1} = c = bs^{-1}$. Rearranging yields $ar^{-1}s = b$. This is a contradiction, as $r^{-1}s$ is a unit, and by assumption a and b were not associates. Thus R is not a principal ideal domain. $\square$

Problem 2 (S23, P3). Suppose R is a PID. Prove that an ideal $I \subset R$ is maximal if and only if $I = (p)$ for a prime $p \in R$.

Proof. Let I be maximal. Since R is a PID, $I = (x)$ for some $x \in R$. If $x = ab$, then $(x) \subseteq (a)$. Hence $(a) = (x)$ or $(a) = R$ by the maximality of (x) . This means that a is a unit or an associate of x . It must be the case that x is irreducible. In a PID, irreducible elements are prime, so x is a prime element and $I = (x) = (p)$ for some prime p .

($\Leftarrow$) Let (m) be any ideal containing (p) . Then $p \in (m)$, so $p = rm$ for some $r \in R$. Since (p) is a prime ideal and $rm \in (p)$, either $r \in (p)$ or $m \in (p)$. If $m \in (p)$ then $(m) = (p)$. If $r \in (p)$, then $r = ps$ for some $s \in \mathbf{Z}$. It follows that $p = rm = psm$, hence $sm = 1$. Since m is a unit, $(m) = R$. Thus (p) is maximal. $\square$

Problem 3 (F17, P3). Prove that every Euclidean domain is a principal ideal domain.

Proof. Let R be a Euclidean domain. Let $I \subset R$ be any ideal. Pick $x \in I$ with minimal norm (by the Well-ordering principle, such an x exists since $\{|r| \mid r \in I\} \subset \mathbf{N}$). Clearly $(x) \subseteq I$, as I is closed under multiplication by elements of R . Let $y \in I$. Write $y = xq + r$, where $q, r \in \mathbf{Z}$ such that $|r| < |x|$ or $r = 0$. Since x was chosen to have minimal norm, it must be the case that $r = 0$. Hence $y = xq$, which gives $y \in (x)$. Thus $I = (x)$, establishing R as a PID. $\square$

Problem 4 (W22, P4). Let $\mathbf{Z}_2 = \{0, 1\}$ be the field of 2 elements. The quotient ring $\mathbf{Z}_2[x]/(x^3 + x + 1)$ be the natural projection.

- (a) Write down a set of 8 distinct coset representatives for the elements of this field.
- (b) Determine the multiplicative inverse of $\pi(x)$ in terms of your coset representatives.

Proof. (a) Note that the coset representatives will be of the form $ax^2 + bx + c$ for $a, b, c \in \{0, 1\}$. The coset representatives for the elements of this field are:

$$\{[0], [1], [x], [x + 1], [x^2], [x^2 + 1], [x^2 + x], [x^2 + x + 1]\}.$$

(b) Note that $[x^3 + x + 1] = [0]$, hence $[x^3] = [-x - 1] = [x + 1]$. It follows that $[x][x^2 + 1] = [x^3 + x] = [1]$, hence the inverse of $[x]$ is $[x^2 + 1]$. $\square$